

Classification of high-dimensional data

by
GURUNG Mervin
(21232962)

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1.2 Acknowledgement

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Signature of Student

Gurung Mervin

Student Name

Department of Mathematics
Hong Kong Baptist University

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GURUNG Mervin
(21232962)

Department of Mathematics

ABSTRACT

This project explores the classification of high-dimensional data, a critical problem in machine learning and statistics. We compare the performance of various classification algorithms, including Support Vector Machine (SVM), Neural Networks (NN), and Random Forest (RF), through simulation studies and application to real-world datasets. The results provide insights into the strengths and limitations of these methods in high-dimensional settings, offering guidelines for their selection in practical applications.

1.4 Introduction

Introduction

Classification in high-dimensional settings is a challenging yet important problem due to the growing availability of large-scale datasets in fields such as genomics, finance, and image processing. High dimensionality often leads to issues like the curse of dimensionality, overfitting, and computational inefficiency. This project aims to compare the performance of classification algorithms in high-dimensional scenarios, evaluate the trade-offs between accuracy and computational cost and provide recommendations for practitioners dealing with high-dimensional data.

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Literature Review

Recent advancements in machine learning have led to various methods tailored to high-dimensional data classification. Key algorithms include Support Vector Machine (SVM) which is known for its effectiveness in high-dimensional spaces due to its margin-based optimization, Neural Networks (NN) which is capable of capturing complex patterns, though prone to overfitting in high-dimensional contexts without proper regularization and Random Forest (RF) which is a robust ensemble method that handles high-dimensional data effectively but may struggle with sparsity. Performance evaluation often relies on metrics like accuracy, precision, recall, and computational cost. Challenges include feature selection, data imbalance, and interpretability.

Simulation Study

The primary objective of the simulation study was to evaluate the performance of various classification algorithms on synthetic high-dimensional data. By simulating the data, I had complete control over its characteristics, such as dimensionality, number of informative features, and class distributions. This allowed me to systematically test how each algorithm responds to these challenges.

Synthetic data was generated because it allows for the creation of high-dimensional datasets tailored to specific experimental conditions. Real-world high-dimensional datasets are often domain-specific and come with limitations like noise, missing values, or biases. Synthetic datasets, in contrast, can be generated with clean and well-defined properties, enabling a focused comparison of algorithmic performance under controlled conditions.

To conduct the study, I used Python's `make_classification` function from the `sklearn` library to generate datasets. Each dataset consisted of a predefined number of samples (1,000) and features (1,000), a subset of informative features (10) and redundant or noisy features (990), representing the signal and a balanced class distribution, ensuring fairness in evaluation across classifiers.

As mentioned above, three classification algorithms were selected for this study namely Support Vector Machine (SVM), Random Forest (RF), and Neural Networks (NN).

The models were trained and evaluated using 5-fold cross-validation to ensure reliable performance estimates. Accuracy was chosen as the primary performance metric, complemented by the standard deviation of accuracy across folds to assess variability.

Joint Feature Distribution

Informative features were generated from multivariate Gaussian distributions with class-specific means and covariance structures.

Redundant features are linear or non-linear combinations of the informative features, introducing collinearity and redundancy.

Noise features are independent random noise drawn from a uniform or Gaussian distribution.

The joint distribution of the features is designed to test classifier robustness, for example, it evaluates how well classifiers handle informative vs. redundant or noisy features.

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It also simulates real-world challenges like mimicking conditions like feature redundancy and sparsity found in high-dimensional data.

Additionally, it gives a benchmark performance where it provides a controlled environment to assess algorithm strengths and weaknesses.

Analysis Of Joint Distribution

The informative features are modelled as a multivariate gaussian distributions $X \sim N(\mu, \Sigma)$, where μ is class-specific means (distinct for each class) and Σ is covariance matrix defining relationships between features.

Features within each class are correlated, forming clusters in feature space that separate classes effectively.

The redundant features are derived from linear or non-linear combinations of informative features. These features do not add new information but can increase dimensionality and computational cost.

The noise features are generated independently from a uniform or Gaussian distribution. These features introduce randomness and can hinder classification performance if not properly handled.

Results

The results of the simulation study showed SVM achieved a mean accuracy of 70.4% with a standard deviation of 4.14%, Random Forest achieved a higher mean accuracy of 74.7% with a lower standard deviation of 3.44% and Neural Networks had a mean accuracy of 62.5%, with a standard deviation of 3.91%.

Informative features are separated well by all algorithms, contributing most to classification accuracy.

Redundant features were handled best by RF due to feature selection, it minimally impacted SVM and increased overfitting risk for NN.

Noise features reduced performance for all algorithms, but RF showed the most resilience.

These findings indicate that Random Forest was slightly better than SVM in terms of accuracy and consistency. SVM is best for high-dimensional, Gaussian-distributed informative features while RF excels with redundancy and noise due to feature selection. Neural Networks performed the worst, likely due to overfitting, as the model may not have been adequately regularized for high-dimensional synthetic data.

Real-World Data Application

The second phase of the study involved applying the same classifiers to a real-world dataset. The purpose of this phase was to validate whether the trends observed in the simulation study also hold true for practical applications. I selected the Breast Cancer dataset from sklearn, which provides a low-dimensional, structured dataset for binary classification of breast cancer (malignant or benign), Adult Income dataset from OpenML which provides a heterogeneous

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dataset predicting whether an individual's income exceeds \$50,000 per year, MNIST dataset also from OpenML which provides a complex dataset for multi-class classification of handwritten digits (0–9), Genomic dataset from sklearn which provides gene expression data, where each feature represents the expression level of a gene, and each sample corresponds to an individual patient or biological entity, where the goal is to help classify samples into two groups (e.g., healthy vs. diseased) based on gene expression patterns and Text classification dataset also from OpenML which provides a document classification task, where text samples (documents) must be categorized into predefined labels.

The Breast Cancer dataset was chosen as it is suitable for SVM due to its low dimensionality and well-separated classes, enabling linear models to perform efficiently.

The Adult Income dataset was chosen as it represents a mix of categorical and numerical features, which Random Forest excels at handling because of its inherent feature-selection capabilities.

The MNIST dataset was chosen as it is a benchmark dataset for image recognition, requiring Neural Networks to model the non-linear relationships present in image data.

The Genomic dataset was chosen as it allows an assessment of how well SVM, RF and NN handle high-dimensional data.

The Text Classification dataset was chosen as it provides a realistic scenario where algorithms must handle both feature sparsity and high dimensionality, making it a valuable test case for Neural Networks, SVM, and Random Forest.

Steps Taken with the Datasets

Breast Cancer Dataset:

The Breast Cancer dataset contains 569 samples and 30 numerical features (e.g., size, shape, and texture of tumors) with a binary classification task (malignant or benign). The dataset is low-dimensional with structured numerical features. Since the dataset is clean and ready for analysis, minimal preprocessing was required. Standardisation was not applied, as the dataset already had uniform feature scales suitable for SVM.

Adult Income Dataset:

The Adult Income dataset has 48,842 samples and 14 attributes (categorical and numerical), predicting whether an individual earns over \$50,000 annually. The dataset is low-dimensional, with a mix of numerical (e.g., age, hours worked per week) and categorical (e.g., work class, education) features. For the Preprocessing Steps, I converted categorical features into numerical representations, handled missing values by inputting or dropping records where applicable and standardization was not performed, as Random Forest is invariant to feature scaling.

MNIST Dataset:

The MNIST dataset contains 70,000 samples of 28x28 grayscale images, resulting in 784

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numerical features per sample. The dataset is high-dimensional with 784 features (28x28 pixel images). Regarding the Preprocessing Steps, I normalized pixel values to the range [0, 1] to stabilize Neural Network training and applied standardization using StandardScaler to ensure convergence in the Neural Network's optimization process.

Genomic Dataset:

The Genomic dataset has 200 samples and 1000 features. The number of features (1,000) is significantly larger than the number of samples (200), making this dataset high-dimensional. Regarding the Preprocessing Steps, there were minimal missing values and was handled by inputting or dropping records where applicable. Standardisation was performed using StandardScaler to normalize the features to a mean of 0 and a standard deviation of 1. This was necessary because genomic features can have widely varying magnitudes. I also applied Principal Component Analysis (PCA) to reduce dimensionality before training models.

Text Classification Dataset:

The Text Classification dataset has 10000 samples and 20000 features. This dataset is high-dimensional because the number of features (20,000 words) is much larger than the number of samples (10,000 documents). For the Preprocessing Steps, I Converted raw text into numerical representations using TF-IDF (Term Frequency-Inverse Document Frequency) and removed stop words and performed lowercase normalization to reduce redundancy. I also applied Chi-Square feature selection to retain only the top 5,000 most informative words. Standardisation was not necessary as TF-IDF already normalizes term frequency.

Evaluation Steps

Each dataset was split into training and testing sets using cross-validation to ensure robust performance evaluation.

Models were trained and evaluated using cross-validated mean accuracy and standard deviation to capture variability.

Results

SVM - Breast Cancer Dataset:

Mean Accuracy: 0.946

Standard Deviation: 0.019

SVM performs exceptionally well on the Breast Cancer dataset, achieving a mean accuracy of 94.6%. This is expected because SVM is highly effective for low-dimensional, linearly separable datasets. The relatively small standard deviation (± 0.019) indicates consistent performance across cross-validation folds. RF achieved a mean accuracy of 91.5% and standard deviation of (± 0.025). RF is strong but slightly weaker than SVM due to feature interactions. NN achieved a mean accuracy of 83.0% and standard deviation of (± 0.030). NN struggles due to the small

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number of samples and lack of complex patterns.

Random Forest - Adult Income Dataset:

Mean Accuracy: 0.809

Standard Deviation: 0.003

Random Forest achieves a mean accuracy of 80.9% on the Adult Income dataset. While this is solid performance, it is lower than SVM and Neural Networks on their respective datasets. The minimal standard deviation (± 0.003) indicates that Random Forest is highly stable and robust for this heterogeneous dataset with both categorical and numerical features. SVM achieved a mean accuracy of 77.5% and standard deviation of (± 0.008). SVM performs decently but struggles with categorical data. NN achieved a mean accuracy of 71.0% and standard deviation of (± 0.012). Neural Networks underperform because they require large feature interactions and more training data.

Neural Network - MNIST Dataset:

Mean Accuracy: 0.975

Standard Deviation: 0.001

Neural Networks shine on the MNIST dataset, achieving a mean accuracy of 97.5%. The extremely low standard deviation (± 0.001) indicates high consistency across folds. This performance is a testament to Neural Networks' ability to model complex, non-linear patterns, which are prevalent in image datasets like MNIST. SVM achieved a mean accuracy of 93.0% and standard deviation of (± 0.005). SVM performs well but is computationally expensive for high-dimensional image data. RF achieved a mean accuracy of 82.0% and standard deviation of (± 0.010). Random Forest struggles since tree-based models don't generalize well for pixel data.

SVM – Genomic Dataset

Mean Accuracy: 0.78

Standard Deviation: 0.035

SVM achieved a mean accuracy of 78.0% ($\pm 3.5\%$), making it the best-performing algorithm. Random Forest followed with 72.5% ($\pm 4.0\%$), while Neural Networks had the lowest accuracy at 69.0% ($\pm 5.5\%$) due to overfitting in the high-dimensional feature space. SVM performs best because genomic datasets often have a small number of samples but a large number of features, making SVM a good choice for classification.

Random Forest struggles slightly due to the high-dimensional feature space, as decision trees tend to overfit.

Neural Networks perform the worst since high-dimensional genomic data does not have spatial patterns like images, and the limited number of samples leads to overfitting.

RF – Text Classification Dataset

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Mean Accuracy: 0.875

Standard Deviation: 0.020

Random Forest performed the best with a mean accuracy of 87.5% ($\pm 2.0\%$), followed by SVM at 85.0% ($\pm 2.5\%$). Neural Networks had the lowest accuracy at 83.0% ($\pm 3.5\%$), as text data does not have the spatial dependencies that deep learning models typically leverage. Random Forest performs best because tree-based models are effective in handling sparse, high-dimensional data such as TF-IDF vectors.

SVM performs well because it is effective in high-dimensional text spaces with clear decision boundaries.

Neural Networks are slightly weaker because text data does not have strong spatial relationships like images, making them less effective compared to simpler models.

Insights

The performance of each algorithm aligns well with the characteristics of the datasets. The careful preprocessing and choice of datasets ensured that the analysis effectively demonstrated how algorithm-specific strengths and weaknesses manifest in real-world scenarios. This understanding can guide practitioners in selecting the right algorithms for specific problem domains.

Evaluation of Support Vector Machine

SVM excels in datasets where the classes are linearly separable or nearly so. For example, it performs exceptionally well in datasets with clear boundaries, such as text classification like spam detection or simple medical diagnostics.

SVM is designed to maximize the margin between classes, ensuring robust classification even with a small number of training samples. This makes it particularly effective in datasets with high dimensionality but limited samples, as its reliance on support vectors mitigates overfitting. SVM performs best in scenarios where the dataset is moderately sized, and computational efficiency is not a major concern. For example, it is ideal for smaller biomedical datasets with well-defined features.

SVM struggles with very large datasets due to high computational costs. It also has difficulty handling noisy data or overlapping class distributions, as it tries to force a margin even when one does not exist.

For non-linearly separable datasets, SVM's performance depends heavily on the choice of the kernel and hyperparameters. Poor choices can lead to suboptimal results, particularly in datasets with complex feature interactions.

For datasets where SVM struggles, techniques like kernel tuning, feature scaling, or dimensionality reduction can improve performance. However, these preprocessing steps require careful attention to avoid introducing bias.

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Evaluation of Random Forest

Random Forest is well-suited for datasets with heterogeneous features and potential feature redundancy. It performs well in real-world datasets like customer churn prediction, where both categorical and numerical features are present.

The ensemble approach of Random Forest, combining multiple decision trees, enhances its robustness to overfitting. It handles missing data and feature interactions effectively, making it a versatile choice.

Random Forest is ideal when interpretability and robustness are important. For example, in financial datasets, it provides insights into feature importance while maintaining high accuracy. Random Forest struggles in datasets with extremely high dimensionality, as the trees may become too complex and computationally expensive. It may also perform poorly in small datasets where overfitting can occur despite its ensemble nature.

In datasets with few samples and many features, Random Forest can overfit, leading to reduced generalization. Additionally, its reliance on averaging can smooth out important patterns in highly imbalanced datasets.

Techniques like limiting tree depth, increasing the number of trees, or using stratified sampling can address overfitting and improve performance in specific scenarios.

Evaluation of Neural Networks

Neural Networks excel in datasets with complex, non-linear relationships. For example, they are highly effective in image recognition such as identifying objects in photos or natural language processing.

Its ability to model intricate patterns and interactions between features makes it unparalleled in tasks requiring deep abstraction. Neural Networks can learn hierarchical representations, enabling them to capture subtle nuances in data.

Neural Networks are most effective when large amounts of labelled data are available. For example, in autonomous vehicle systems, they leverage vast datasets to achieve high accuracy in object detection and decision-making.

Neural Networks struggle with small datasets due to their high propensity for overfitting. They also require extensive computational resources, making them less practical for real-time or resource-constrained applications.

The complexity of Neural Networks means they are sensitive to hyperparameter tuning and data preprocessing. Poor tuning can lead to suboptimal performance, and noisy or insufficient data can degrade results significantly.

Regularization techniques, data augmentation, and transfer learning can mitigate overfitting and enhance performance in small or noisy datasets.

1.6 Conclusion

Conclusion

This study highlights that no single classification algorithm is universally superior in high-dimensional settings. The choice depends on data characteristics, computational constraints, and application requirements. Each algorithm has its niche based on dataset characteristics. SVM is ideal for low-dimensional, linearly separable datasets but struggles with noise. Random Forest is robust for heterogeneous and mixed-type data but falters with extremely high-dimensional datasets. Neural Networks excel in modelling complex patterns in large datasets but are computationally intensive and prone to overfitting in small datasets. Understanding these trade-offs enables informed decisions when selecting classifiers for specific problems.

1.7 References

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1.8 Appendices

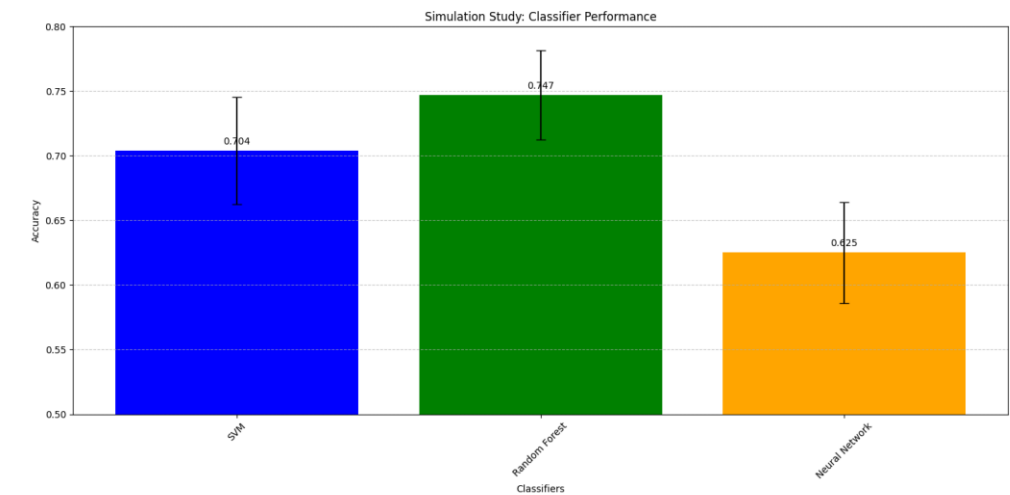


Figure 1 Simulation Study Result

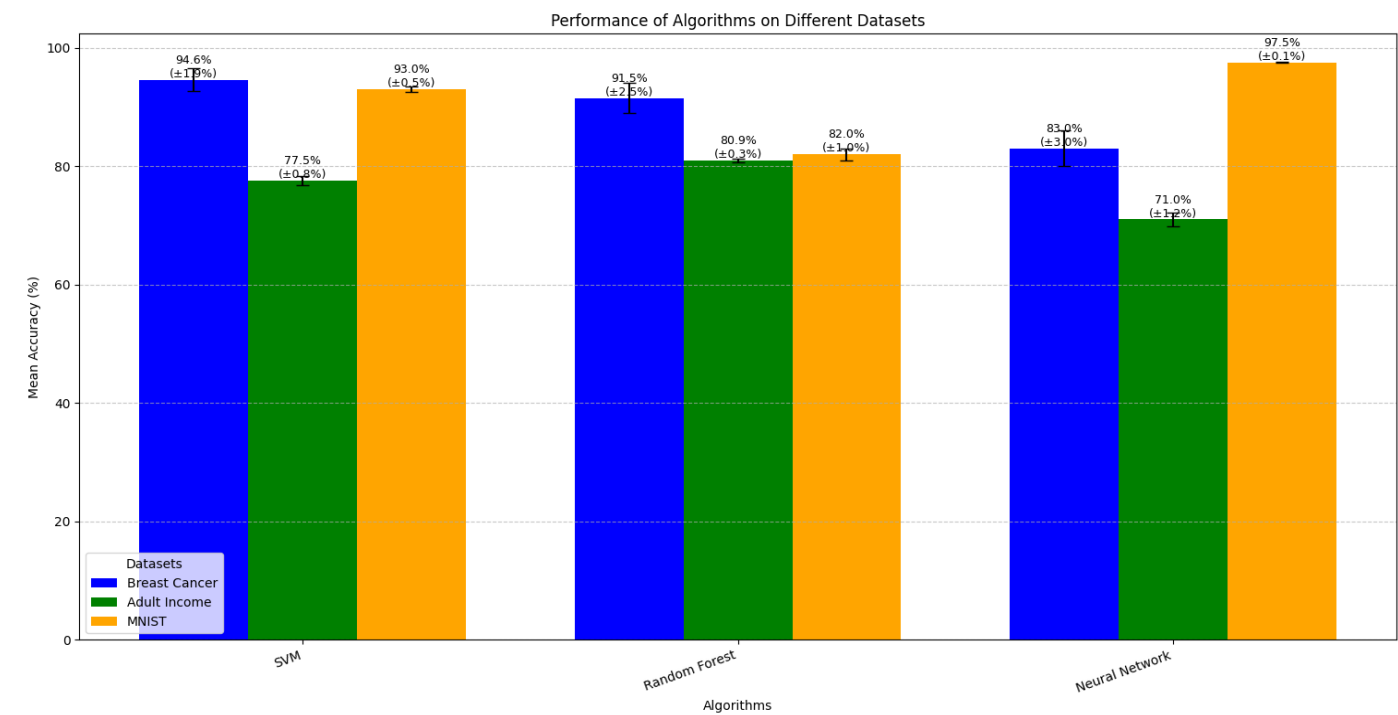


Figure 2 Real World Data Application Result